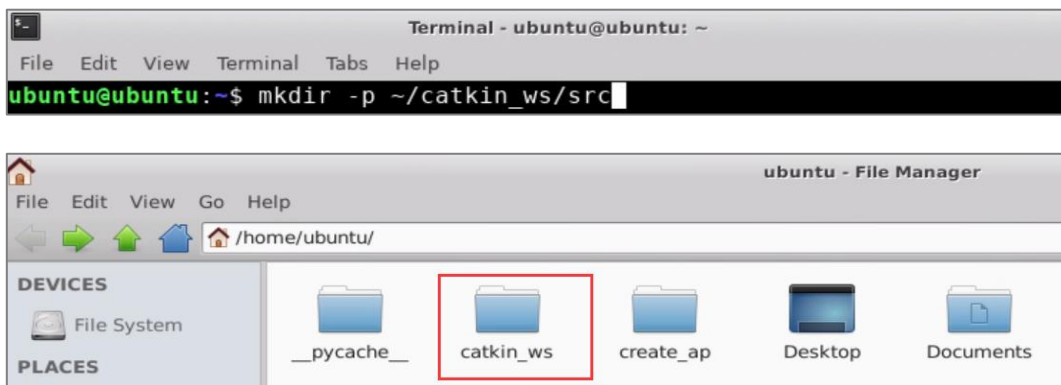


第 4 课 创建工作空间与功能包

消息和服务的创建、发布器和订阅器的编写、服务端和客户端的编写都是基于Ros功能包进行操作的，因此在进行上述操作前，需要先创建工作空间及功能包。

1.工作环境的创建

- 1) 打开命令行终端。
- 2) 输入指令“**mkdir -p ~/catkin_ws/src**”，并按下回车，在指定目录下创建一个名为catkin_ws的工作空间。



2.ROS功能包的创建

- 1) 输入指令“**cd ~/catkin_ws/src**”，并按下回车，进入catkin_ws的源文件空间（src）目录。



2) 输入指令“**catkin_create_pkg beginner_hiwonder std_msgs rospy roscpp**”，并按下回车，创建一个名为“**beginner_hiwonder**”的功能包，添加依赖关系(msgs、rospy、roscpp)。

```
Terminal - ubuntu@ubuntu: ~/catkin_ws/src
File Edit View Terminal Tabs Help
ubuntu@ubuntu:~/catkin_ws/src$ catkin_create_pkg beginner_hiwonder std_msgs rospy roscpp
Created file beginner_hiwonder/package.xml
Created file beginner_hiwonder/CMakeLists.txt
Created folder beginner_hiwonder/include/beginner_hiwonder
Created folder beginner_hiwonder/src
Successfully created files in /home/ubuntu/catkin_ws/src/beginner_hiwonder. Please adjust the values
in package.xml.
```

3) 输入指令“**cd ~/catkin_ws**”，并按下回车，返回catkin工作空间。

```
Terminal - ubuntu@ubuntu: ~/catkin_ws/src/beginner_hiwonder
File Edit View Terminal Tabs Help
ubuntu@ubuntu:~/catkin_ws/src/beginner_hiwonder$ cd ~/catkin_ws
```

4) 输入指令“**catkin_make**”，并按下回车，完成catkin工作区中功能包的构建。

```
Terminal - ubuntu@ubuntu: ~/catkin_ws
File Edit View Terminal Tabs Help
ubuntu@ubuntu:~/catkin_ws$ catkin_make

-- traversing 1 packages in topological order:
--   - beginner_hiwonder
-- +++ processing catkin package: 'beginner_hiwonder'
-- ==> add_subdirectory(beginner_hiwonder)
-- Configuring done
-- Generating done
-- Build files have been written to: /home/ubuntu/catkin_ws/build
#####
##### Running command: "make -j4 -l4" in "/home/ubuntu/catkin_ws/build"
#####
```

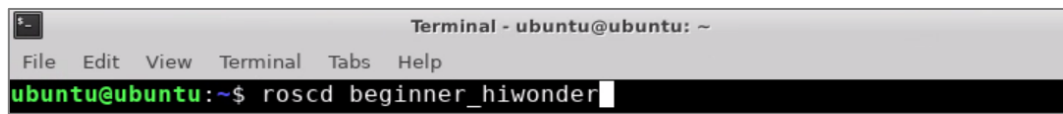
5) 输入指令“**echo "source ~/catkin_ws/devel/setup.bash" >> ~/.bashrc**”，将catkin_ws工作空间添加到ROS环境中，即完成功能包的创建。

```
Terminal - ubuntu@ubuntu: ~
File Edit View Terminal Tabs Help
ubuntu@ubuntu:~$ echo "source ~/catkin_ws/devel/setup.bash" >> ~/.bashrc
```

6) 输入指令“**source ~/.bashrc**”，并按下回车，即完成路径的添加。

```
Terminal - ubuntu@ubuntu: ~
File Edit View Terminal Tabs Help
ubuntu@ubuntu:~$ source ~/.bashrc
```

7) 输入进入功能包所在目录的指令“**roscd beginner_hiwonder**”，并按下回车，验证功能包是否创建成功。

A screenshot of a terminal window titled "Terminal - ubuntu@ubuntu: ~". The terminal has a menu bar with "File", "Edit", "View", "Terminal", "Tabs", and "Help". The command prompt shows "ubuntu@ubuntu:~\$ roscd beginner_hiwonder" with a cursor at the end of the command.

```
Terminal - ubuntu@ubuntu: ~
File Edit View Terminal Tabs Help
ubuntu@ubuntu:~$ roscd beginner_hiwonder
```